

Ham : "Lunch is ready"
 Spam : "Winner! Free Lunch"
 Ham : "Free is good"

"1" Prior $P(\text{class})$

Total messages = 3

Spam = 1, $P(\text{Spam}) = 1/3 \approx 0.33$

Ham = 2, $P(\text{Ham}) = 2/3 \approx 0.67$

(1)
 $P(\text{word}|\text{class}) = \frac{\text{count of words in class} + \alpha}{\text{total words in class} + \alpha \times V}$
 $\alpha = \text{Laplace Smoothing Factor}$
 $V = \text{Vocabulary Size}$
 apply Laplace Smoothing

"2" Likelihoods $P(\text{word}|\text{class})$

↳ we count how many times a word appears in a specific class divided by total number of words in that class.

Vocabulary (unique words in all messages)

$V = \text{free, good, is, lunch, ready, winner}$

$\text{Size}(V) = 6$

Word	Count in Spam	Count in Ham	$P(\text{word} \text{Spam})$	$P(\text{word} \text{Ham})$
free	1	1	$\frac{1+1}{3+6} = 0.22$	$\frac{1+1}{6+6} = 0.16$
winner	1	0	$\frac{1+1}{3+6} = 0.22$	$\frac{0+1}{6+6} = 0.08$
lunch	1	1	$\frac{1+1}{3+6} = 0.22$	$\frac{1+1}{6+6} = 0.16$

(2)

classify a new message
"Free Winner"

Laplace Smoothing
ensures every word gets
a tiny, tiny scale of
prob in every class.

Score for Spam:

$$P(\text{Spam}) \cdot P(\text{Free}|\text{Spam}) \cdot P(\text{winner}|\text{Spam})$$

$$0.33 \times 0.22 \times 0.22 = 0.0159$$

Score for Ham:

$$P(\text{Ham}) \cdot P(\text{Free}|\text{Ham}) \cdot P(\text{winner}|\text{Ham})$$

$$0.67 \times 0.16 \times 0.08 = 0.0085$$

$$\text{Spam Score (0.0159)} > \text{Ham Score (0.0085)}$$

"Free winner" → Spam

Actual Prob

$$\frac{0.0159}{0.0159 + 0.0085} = 65.1\%$$

input message
"winner! lunch is
ready"

→ Why The Smoothing (Laplace, adding a noise).

Imagine "winner" appeared in spam but not in Ham

without smoothing $P(\text{winner}|\text{Ham}) = 0$

she for calculating final score for message "winner! lunch is ready".

$$0.08 \times 0.16 \times 0 = 0$$